## NAME

**mbtrnpp** − Real-time preprocessor that reads multibeam sonar data, applies automated cleaning and downsampling, and passes the resulting bathymetry to a terrain relative navigation (TRN) process.

## VERSION

Version 5.0

## SYNOPSIS

**mbtrnpp**
[--verbose
--help
--config=path
--log-directory=path
--input=datalist|file|socket_definition
--output=file|socket
--format=format
--platform-file=file
--platform-target-sensor=sensor_id
--tide-model=file
--swath-width=value
--soundings=value
--median-filter=threshold/nx/ny
--statsec=d.d
--statflags=<MSF_STATUS|MSF_EVENT|MSF_ASTAT|MSF_PSTAT|MSF_READER>
--delay=n
--mb-out=mb1svr[:host:port]/mb1/file[:path]/reson/nomb1/noreson/nombsvr/nombtrnpp
--mbhbn=n
--mbhbt=d.d
--trn-en
--trn-dev=s
--use-proj[=bool]
--projection=projection_id
--trn-crs=s
--trn-utm=zone
--trn-map=path
--trn-cfg=path
--trn-par=path
--trn-mid=id
--trn-mtype=n
--trn-sensor-type=n
--trn-ftype=n
--trn-fgrade=n
--trn-freinit=n
--trn-mweight=n
--trn-ncov=value
--trn-nerr=value
--trn-ecov=value
--trn-eerr=value
--trn-decn=n
--trn-decs=d.d
--covariance-magnitude-max=value
--convergence-repeat-min=n

--reinit-search=reinit_search_xy/reinit_search_z
--reinit-gain[=bool]
--reinit-file[=bool]
--reinit-xyoffset=xyoffset_max
--reinit-zoffset=zoffset_min/zoffset_max
--random-offset
--trn-out=trnsvr[:host:port]/trnusvr[:host:port]/trnumsvr[:group:port:ttl]/trnu/trnub/sout/serr/debug/notrnsvr/notrnusvr/notrnumsvr
--trnhbt=d.d
--trnuhbt=d.d]

## DESCRIPTION

**mbtrnpp** is a command line program, normally run unattended aboard an autonomous underwater vehicle (AUV) or other survey platform, that reads multibeam bathymetry in real time (from a datalist, a single swath data file, or a live network socket), applies automated cleaning, swath-width limiting, and sounding decimation, and then passes the resulting bathymetry soundings on to a terrain relative navigation (TRN) engine embedded in the program. **mbtrnpp** is one of the **MB-System** utilities in the **mbtrnutils** source directory; it is an optional program, built only when **MB-System** is configured with TRN support enabled (**--enable-mbtrn** with autotools, or **-DbuildTRN=ON** with **CMake**, both of which are the default), and further requires TRN support (**--enable-mbtnav** or **-DbuildTRNLCM**-related **CMake** options) to actually run TRN rather than simply relaying bathymetry.

**mbtrnpp** has no interactive display; it is configured entirely through command line options and, optionally, a configuration file specified with **--config**. The order of precedence for setting an option is (highest to lowest) command line options, configuration file options, and compiled-in defaults. Because **mbtrnpp** is normally run unattended and continuously, most deployments use a shell wrapper that first sources an environment file (setting variables such as **TRN_HOST**, **TRN_MAPFILES**, **TRN_DATAFILES**, and **TRN_LOGFILES** that are referenced by mnemonic in the configuration file and by **mbtrnpp** itself), and then invokes **mbtrnpp** with **--config** pointing to a configuration file tailored to the vehicle and mission. The **--help** option is useful even in normal operation, since **mbtrnpp** prints its default, configuration-file, and final (command-line-overridden) settings to stderr as it starts up, before exiting without reading any data; this allows a configuration to be checked without actually running a survey.

Input bathymetry may come from a recursive **MB-System** datalist or a single swath data file (in which case **--format** normally must also be given), read exactly as with other **MB-System** utilities, or from a live data socket (e.g. a Reson/Teledyne multibeam sonar or 7k Center, specified with an **--input** value beginning *socket:*) for real-time operation. As pings are read, **mbtrnpp** applies an automated median filter across and along track to flag anomalous soundings, limits the swath to the width set by **--swath-width**, and decimates each ping to the number of soundings set by **--soundings**. The processed soundings are packaged into MB1 records, which may be logged to a binary file, served to network clients, and/or passed directly to the embedded TRN instance for terrain-relative position estimation (when TRN is enabled with **--trn-en** and **MB-System** was built with TRN navigation support). TRN position estimates and updates may in turn be logged and/or served to network clients using the **--trn-out** option.

**mbtrnpp** writes several kinds of log files (an MB-System processing log, an mbtrnpp session log, MB1 server and TRN server/update session logs, the MB1 binary record log, and, if TRN is enabled, a TRN instance log directory) into the directory set by **--log-directory**; a symbolic link named **mbtrnpp-latest** pointing at this directory is created (or replaced) each time **mbtrnpp** starts. See the **README-mbtrnpp.md** file distributed with **MB-System** in **src/mbtrn/tools/cfg** for a complete description of the environment and configuration files, the network output specifiers, log file naming, and troubleshooting tips.

## MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Kent Headley
Monterey Bay Aquarium Research Institute
Christian dos Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

## OPTIONS

**−-verbose**

This option increases the verbosity of **mbtrnpp**, which means that more information than by default is output to the stderr stream of the shell. Negative values (e.g. **--verbose=-2**) also select the printing of the program name and **MB-System** version banner at startup.

**−-help**

This option causes **mbtrnpp** to print a description of what it does and a summary of the command line options, and then exit without reading any data. Because **mbtrnpp** always prints its default, configuration-file, and final settings to stderr before checking for **--help**, this option is a convenient way to verify a configuration (including one loaded with **--config**) without starting a survey.

**−-config=path**

This option specifies a configuration file containing any of the **mbtrnpp** command line options, one *key***=***value* pair per line (using the same option names as the command line, without the leading **--**). Configuration file options override the compiled-in defaults, and are in turn overridden by the same option given explicitly on the command line. The **--config** option itself is recognized before any other command line processing, so a configuration file may be combined with command-line overrides in a single invocation. See the annotated example configuration file **src/mbtrn/tools/cfg/mbtrnpp-cfg.example** distributed with **MB-System**.

**−-log-directory=path**

This option sets the directory into which **mbtrnpp** writes its log files (the **MB-System** processing log, the mbtrnpp and MB1/TRN server session logs, the MB1 binary record log, and, if TRN is enabled, the TRN instance log directory). The directory is created if it does not already exist. The default is the current working directory (**.**). A symbolic link named **mbtrnpp-latest** pointing at this directory is created (or replaced) each time **mbtrnpp** starts.

**−-input=datalist|file|socket_definition**

This option specifies the source of input bathymetry data, exactly as the **--input** (**-I**) option does in other **MB-System** utilities, with the addition of live socket input. If the value does not begin with *socket:*, it is read as an **MB-System** recursive datalist (in which case **--format** is not required, since each listed file names its own format) or as a single swath data file (in which case **--format** must also be given). If the value begins *socket:*, the remainder is a socket definition string (e.g. identifying a Reson/Teledyne sonar or 7k Center host and port) used to read multibeam pings from a live network connection instead of a file; **--format** must be set to the format ID appropriate for the socket data stream. The default input is **stdin**.

**−-output=file|socket**

This option enables output of processed MB1 records to a binary file and/or a network socket server. A value of *file* enables logging to a file with the default **MB-System**-generated name; a value of the form *file:path* logs to *path* instead. A value of *socket* enables the MB1 socket server using its default host and port; a value of the form *socket:host***:***port* overrides the host and/or port. Multiple comma-separated values may be given. This option predates, and is functionally a subset of, the more general

**--mb-out** option; new configurations should prefer **--mb-out**.

**−-format=format**
: This option sets the **MB-System** format ID of the input data, exactly as the **--format** (**-F**) option does in other **MB-System** utilities. It is required when **--input** names a single file or a live socket; it is not needed when **--input** names a datalist, since each file listed in the datalist specifies its own format.

**−-platform-file=file**
: This option specifies an **MB-System** platform definition file describing the sensor offsets and orientation of the survey platform (vehicle), exactly as used by **mbpreprocess** and other **MB-System** utilities that support platform files. See the **MB-System** manual page for a description of the platform file format.

**−-platform-target-sensor=sensor_id**
: This option selects, by ID, the sensor within the platform file (given by **--platform-file**) whose offsets are applied to the bathymetric soundings.

**−-tide-model=file**
: This option specifies a tide model file used to remove tidal variation from the input bathymetry, following the same tide model file format used by **mbotps** and **mbprocess** elsewhere in **MB-System**.

**−-swath-width=value**
: This option sets the width of the sonar swath, in decimal degrees measured from nadir, that is retained from each ping; soundings outside this swath are excluded from output. The default is 90.0 (the full swath).

**−-soundings=value**
: This option sets the number of soundings output per ping after decimation. Soundings are selected evenly across the retained swath. The default is 11.

**−-median-filter=threshold/nx/ny**
: This option enables an automated median filter applied to the bathymetry before decimation, and sets its parameters: *threshold* is the fractional deviation from the local median depth (relative to depth) beyond which a sounding is flagged bad, and *nx* and *ny* are respectively the number of soundings across track and the number of pings along track included in the local neighborhood used to compute the median. If this option is not given, the median filter is disabled.

**−-statsec=d.d**
: This option sets the interval, in seconds, at which **mbtrnpp** logs periodic processing statistics (selected by **--statflags**).

**−-statflags=<MSF_STATUS|MSF_EVENT|MSF_ASTAT|MSF_PSTAT|MSF_READER>**
: This option selects which categories of processing statistics **mbtrnpp** logs, as a **|**- (or **:**-) delimited list of flag names: *MSF_STATUS* enables status counters, *MSF_EVENT* enables event/error counters, *MSF_ASTAT* enables aggregate (cumulative) statistics, *MSF_PSTAT* enables periodic (interval) statistics, and *MSF_READER* enables statistics from the socket data reader. The default is *MSF_STATUS*|*MSF_EVENT*|*MSF_ASTAT*|*MSF_PSTAT*.

**−-delay=n**
: This option sets a delay, in milliseconds, added to the main MB1 processing loop on each iteration; it may be used to throttle processing when input data arrive faster than downstream consumers (TRN, network clients) can keep up. The default is 0 (no delay).

**−-mb-out=mb1svr[:host:port]/mb1/file[:path]/reson/nomb1/noreson/nombsvr/nombtrnpp**
: This option configures MB1 (processed bathymetry) output, as a **,**-delimited list of tokens: *mb1svr* (optionally followed by **:***host***:***port*) enables the MB1 network server; *mb1* enables binary MB1 record logging using the default log file name; *file* (optionally followed by **:***path*) enables binary MB1 record logging, to *path* if given; *reson* enables logging of the raw Reson/Teledyne S7K data frames; and the *no*-prefixed tokens (*nomb1*, *noreson*, *nombsvr*, *nombtrnpp*) disable the corresponding output (*nombtrnpp* disables the mbtrnpp message log itself, which is not recommended). Tokens are

processed in order, so a later token can override an earlier one on the same command line.

**−-mbhbn=n**
This option sets the number of missed heartbeats the MB1 server tolerates from a client before dropping the connection.

**−-mbhbt=d.d**
This option sets the heartbeat timeout, in seconds, for the MB1 server.

**−-trn-en**
This option enables terrain relative navigation processing using the embedded TRN instance; **mbtrnpp** must have been built with TRN navigation support (**--enable-mbtnav**, or the corresponding **CMake** option) for TRN processing to actually occur. When disabled, **mbtrnpp** still reads, cleans, and outputs bathymetry, but does not attempt terrain navigation. TRN is enabled by default; use **--trn-en=N** (or **=0**) to disable it.

**−-trn-dev=s**
This option selects the sonar device type used to interpret live socket input, by mnemonic (e.g. *7125_200* for a Seabat 7125 at 200 kHz, *7125_400* for a Seabat 7125 at 400 kHz, or *T50* for a T50-S/T50-R). The mnemonic is translated internally to the device and system enumerator values required to subscribe to the sonar's data stream; see **r7kc.h** in the **MB-System** source for the full mapping.

**−-use-proj[=bool]**
This option enables (the default) or disables use of the **PROJ** cartographic projection library to convert geographic coordinates to the projected coordinate system used internally by TRN, instead of a simple UTM calculation.

**−-projection=projection_id**
This option sets the numeric **PROJ** projection ID used when **--use-proj** is enabled.

**−-trn-crs=s**
This option sets the **PROJ** coordinate reference system string used when **--use-proj** is enabled, in place of **--projection**.

**−-trn-utm=zone**
This option sets the UTM zone (an integer from 1 to 60) used for TRN processing when **--use-proj** is disabled (e.g. zone 9 for the axial seamount test site, zone 10 for Monterey Bay).

**−-trn-map=path**
This option specifies the path to the terrain reference map used by the TRN server; it may be a single map file or a directory of tiled map files. A terrain map is required for TRN processing to converge.

**−-trn-cfg=path**
This option specifies the path to the TRN configuration file, which sets per-vehicle TRN filter and sensor specification parameters. A TRN configuration file is required for TRN processing.

**−-trn-par=path**
This option specifies the path to a TRN particle filter configuration file, used only when the TRN particle filter is selected (see **--trn-ftype**).

**−-trn-mid=id**
This option sets the TRN mission ID string used in naming the TRN server's own log files and log directory.

**−-trn-mtype=n**
This option sets the TRN terrain map type: *1* selects a DEM (elevation grid) map, and *2* selects a bottom-object (BO) map.

**−-trn-sensor-type=n**
This option sets the TRN sensor type identifying the source of bathymetry supplied to TRN; the default corresponds to a generic **MB-System** (**MB1**) multibeam source.

**−-trn-ftype=n**
This option sets the TRN estimation filter type: *0* disables filtering, *1* selects a point-mass filter, *2* selects a particle filter, and *3* selects a filter bank.

**−-trn-fgrade=n**
This option sets the TRN filter grade: *0* selects a low-grade filter configuration, and *1* (the default) selects a high-grade filter configuration.

**−-trn-freinit=n**
This option enables (*1*, the default) or disables (*0*) automatic reinitialization of the TRN estimation filter, e.g. after divergence.

**−-trn-mweight=n**
This option selects the TRN modified-weighting scheme used during filtering: *0* applies no weighting modification, *1* applies the original alpha modification, *2* applies crossbeam weighting with the alpha modification, *3* applies subcloud weighting with the original alpha modification, and *4* (the default) applies subcloud weighting with modified NIS (normalized innovation squared) always enabled.

**−-trn-ncov=value**
This option sets the TRN convergence criterion for northing covariance; a TRN solution is not considered converged unless its northing covariance is below this value.

**−-trn-nerr=value**
This option sets the TRN convergence criterion for northing error.

**−-trn-ecov=value**
This option sets the TRN convergence criterion for easting covariance.

**−-trn-eerr=value**
This option sets the TRN convergence criterion for easting error.

**−-trn-decn=n**
This option sets the TRN update decimation modulus: a TRN position update is computed and issued only for every *n*th processed MB1 sounding record. The default is 0 (update on every record).

**−-trn-decs=d.d**
This option sets a minimum time interval, in seconds, between TRN position updates, as a decimation criterion alternative (or in addition) to **--trn-decn**.

**−-covariance-magnitude-max=value**
This option sets the maximum TRN covariance magnitude below which a TRN solution is considered a candidate for stable convergence; smaller values impose a stricter convergence requirement. A value around 10.0 is typical for 1-meter-scale mapping.

**−-convergence-repeat-min=n**
This option sets the number of consecutive TRN update cycles that must meet the convergence criteria (**--covariance-magnitude-max** and related limits) before the TRN solution is declared stably converged. A value around 400 is typical for 1-meter-scale mapping.

**−-reinit-search=reinit_search_xy/reinit_search_z**
This option sets the size, in meters, of the horizontal (*reinit_search_xy*) and vertical (*reinit_search_z*) search area used by TRN when reinitializing its position estimate. The defaults are 60 m horizontal and 5 m vertical.

**−-reinit-gain[=bool]**
This option enables or disables gating TRN reinitialization using the sonar’s transmit gain setting (e.g. to avoid reinitializing on pings taken with an atypical gain). Disabled by default.

**−-reinit-file[=bool]**

This option enables or disables reinitializing TRN whenever a new input file begins while reading from a datalist. Disabled by default; only meaningful when reading from a datalist or file rather than a live socket.

**−-reinit-xyoffset=xyoffset_max**

This option enables reinitialization of TRN whenever the magnitude of its converged horizontal (xy) offset from the vehicle's dead-reckoned position exceeds *xyoffset_max*, in meters.

**−-reinit-zoffset=zoffset_min/zoffset_max**

This option enables reinitialization of TRN whenever the magnitude of its converged vertical (z) offset falls below *zoffset_min* or exceeds *zoffset_max*, in meters.

**−-random-offset**

This option enables injection of a randomized offset into the TRN filter's initial position estimate; this is intended for testing TRN's ability to reconverge after reinitialization, and should not normally be used during an actual survey.

**−-trn-out=trnsvr[:host:port]/trnusvr[:host:port]/trnumsvr[:group:port:ttl]/trnu/trnub/sout/serr/debug/notrnsvr/notrnusvr/notrnumsvr**

This option configures TRN position/update output, as a **,**-delimited list of tokens: *trnsvr* (optionally followed by **:***host***:***port*) enables the TRN TCP server interface; *trnusvr* (optionally followed by **:***host***:***port*) enables the TRN update (TRNU) TCP server; *trnumsvr* (optionally followed by **:***group***:***port***:***ttl*) enables a TRN update UDP multicast server; *trnu* enables ASCII logging of TRN update records, and *trnub* enables binary logging of the same; *sout* and *serr* send TRN output to stdout and stderr, respectively; *debug* enables verbose TRN debug output; and the *no*-prefixed tokens (*notrnsvr*, *notrnusvr*, *notrnumsvr*) disable the corresponding server. The mnemonics **TRN_HOST** and **TRN_GROUP** may be used in place of an explicit *host* or *group*, in which case the corresponding environment variable is substituted at run time.

**−-trnhbt=d.d**

This option sets the heartbeat timeout, in seconds, for the TRN TCP server.

**−-trnuhbt=d.d**

This option sets the heartbeat timeout, in seconds, for the TRN update (TRNU) TCP server.

## EXAMPLES

Suppose you want to check a mission configuration before running a survey, without actually starting to process data. Sourcing the mission's environment file first makes its mnemonic variables (**TRN_HOST**, **TRN_MAPFILES**, and so on) available for substitution in the configuration file:

```
. env/mbtrnpp-Sentry.env
mbtrnpp --config=TRNcfg/mbtrnpp-Sentry.cfg --help
```

**mbtrnpp** prints its default settings, the settings loaded from the configuration file, and the final effective settings to stderr, then exits without reading any data.

To actually run the same configuration as a real-time TRN preprocessor, logging to a mission-specific log directory:

```
. env/mbtrnpp-Sentry.env
mbtrnpp --config=TRNcfg/mbtrnpp-Sentry.cfg \
      --log-directory=/home/mappingauv/logs/mbtrnpp
```

To process a single existing swath file offline (e.g. to test cleaning and TRN convergence against a previously recorded survey) rather than a live sonar socket, overriding the configuration file's **--input** setting:

```
mbtrnpp --config=TRNcfg/mbtrnpp-Sentry.cfg \
      --input=20240510_120000.mb88 --format=88
```

## SEE ALSO

**mbsystem**(1), **mbio**(1), **mbpreprocess**(1), **mbclean**(1), **mbotps**(1)

## BUGS

Unsupervised real-time terrain navigation is inherently harder to debug than the interactive **MB-System** tools it borrows cleaning logic from; when something looks wrong, start by rereading the log files and rechecking the configuration with **--help** before suspecting the code.